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Final year project proposal

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Incubation and Innovation Management System (IIMS)

The **Incubation and Innovation Management System (IIMS)** is a digital platform designed to transform how incubation centers manage programs and support innovation-driven entrepreneurship. It aims to provide a structured, multi-tenant system that automates complex processes such as startup applications, mentorship management, communication, progress tracking, and event scheduling while fostering collaboration and transparency among stakeholders.

The motivation behind IIMS arose from our experience as students at **Bahir Dar University**, where we were actively involved at the **BiTec (BiT Business Incubation and Techno-Entrepreneurship Center)**. While many students had creative ideas, they often struggled to move innovations beyond the concept stage. This was not due to a lack of skill but inefficiencies in managing incubation activities. Startups had difficulty tracking mentorship sessions, receiving timely feedback, and monitoring progress, while administrators struggled to oversee operations effectively.

To understand these challenges, we conducted **requirement gathering and observations** at centers including **BiTec, Orbit Innovation Hub, Entrepreneurship Development Institute (EDI), Ministry of Labour and Skills (MOLS)**, and **innobiz-K**, these visits helped us identify common pain points. A major issue was the **overload and time burden on center managers**, who manually handled applications, mentor assignments, progress tracking, and event organization. This manual approach caused delays, data loss, and miscommunication.

Simultaneously, **startups faced challenges** in tracking application status, managing mentor feedback, and monitoring milestones. They had limited opportunities for structured collaboration and knowledge sharing. **Mentors and investors** also struggled to communicate with startups effectively, as interactions were scattered across emails and spreadsheets. The absence of an integrated **communication and collaboration system** often led to delays or lost information, weakening the incubation process. Together, these factors created a **time-consuming, disconnected, and inefficient** system, limiting the success of incubation programs.

The **IIMS solution** is a **multi-tenant, web-based platform** that allows multiple centers to operate independently while sharing a common infrastructure. Each center will have **customizable modules**, allowing adaptation to specific workflows, programs, and data requirements. University centers may focus on mentorship and project tracking, while government centers may emphasize grants and reporting.

The implemented version of the Incubation and Innovation Management System (IIMS) provides a structured multi-tenant platform that automates core incubation operations. The system supports four main roles **SUPER_ADMIN, TENANT_ADMIN, STARTUP, and MENTOR** each with dedicated dashboards and profile management. The SUPERADMIN manages tenant registration, subscription plans, and feature access control, ensuring proper multi-tenant operation. The TENANT_ADMIN oversees application review, mentor assignment, and startup monitoring. STARTUP users can manage their profiles, track milestones, and submit progress updates, while MENTOR users monitor assigned startups and provide feedback.

The system includes a progress tracking module that manages incubation phases and tasks, along with a mentor assignment management feature for structured guidance. IIMS also integrates communication and collaboration tools, including in-app chat, meeting scheduling, and Google Calendar integration. Automated notifications keep all users informed about updates, deadlines, and meetings. Additionally, a help center management module allows tenants to request support and submit feature inquiries, creating a centralized and efficient incubation management environment.

To further strengthen stakeholder participation and operational efficiency, the next phase of IIMS will expand its role structure beyond the current four roles. New roles: **ALUMNI, INVESTOR, JUDGE, and TENANT_EMPLOYEE** will be introduced to enhance post-incubation engagement, streamline competition management, facilitate investment connections, and improve internal task delegation within incubation centers.

Building on the current implementation, IIMS will be further enhanced with advanced functional modules designed to strengthen the entire incubation and innovation lifecycle. The **Analytics and Reporting module** will provide real-time dashboards, performance indicators, cohort benchmarking, and customizable reports to support data-driven decision-making. An integrated **Alumni Mentorship feature** will reconnect graduated startups into the ecosystem as mentors and collaborators, ensuring continuous knowledge exchange. The **Graduation Management module** will formalize startup exit processes through structured evaluation criteria, readiness assessments, certification generation, and post-incubation tracking.

In addition, a Competition and **Judge Management feature** will support hackathons, pitch days, and innovation challenges by enabling administrators to assign judges, define scoring criteria, and manage evaluations within the platform. Judges will be able to review submissions, score startups, and provide feedback through a dedicated interface, while the system automatically compiles results to ensure fairness and transparency.

The **Content Management module** will allow centers to publish news, opportunities, and resources, while **Landing Page Management** will enable each tenant to manage its own public-facing portal. The **Alumni Management module** will support long-term engagement and impact measurement, and the **Investor Connections feature** will facilitate structured networking between startups and potential investors. Finally, **AI-powered support tools** will introduce intelligent assistance such as automated recommendations and smart progress insights, further enhancing operational efficiency and startup guidance. Together, these extended modules elevate IIMS into a comprehensive, scalable, and innovation-driven incubation management ecosystem.

In conclusion, the **Incubation and Innovation Management System (IIMS)** is more than a software project, it is a solution addressing real challenges. By transforming fragmented manual processes into an integrated platform, IIMS promotes sustainable growth, collaboration, and measurable impact. This project demonstrates how

technology can empower incubation centers, startups, and mentors, turning ideas into tangible results and opportunities.

NannyCare

NannyCare is an innovative digital platform built to reshape childcare in Ethiopia by connecting mothers with skilled, verified, and empowered nannies. It directly responds to one of the country's growing social challenges the lack of structured, trustworthy childcare systems for working mothers and the under-recognized role of caregivers who provide essential family support. Through technology, training, and community engagement, NannyCare brings structure, safety, and dignity to childcare.

For mothers, NannyCare offers a comprehensive solution that goes beyond simple childcare matching. The web app provides easy access to verified nannies and a suite of digital tools for daily parenting support. Mothers can track feeding, sleeping, growth milestones, and vaccination schedules for their children. Wellness tools such as guided breathing exercises, stress management tips, and mood tracking help mothers balance personal well-being with their caregiving responsibilities. The app also enables direct consultations with verified pediatricians, nurses, and midwives, ensuring expert advice is always within reach. Furthermore, NannyCare fosters a sense of belonging by offering a community space where mothers can connect, exchange ideas, share experiences, and form meaningful local support networks.

For nannies, NannyCare is not just a job platform it's a pathway to empowerment and professional growth. The system provides structured online training in childcare fundamentals, hygiene, nutrition, early learning, and first aid. These courses are designed to meet local realities while incorporating global standards of care. Upon

completion, nannies earn digital certificates that validate their expertise and enhance employability. The job board within the platform lists verified opportunities, ensuring transparency and fairness in employment. Beyond job access, nannies receive essential information about fair wages, workers' rights, and career development, enabling them to improve their working conditions and build long-term, sustainable livelihoods.

From a technical perspective, NannyCare is built with a focus on scalability, security, and user experience. The frontend is developed using React.js, ensuring a fast and responsive interface across devices. The backend is powered by Spring Boot, providing a robust framework for API management, data processing, and authentication. PostgreSQL serves as the primary relational database for efficient and secure data storage. Docker containerization enables seamless deployment and scalability.

Beyond its technical innovation, NannyCare's true strength lies in its social impact. The platform lightens the emotional and logistical load on working mothers, helps nannies gain respect and financial stability, and ensures children receive consistent, nurturing, and developmentally supportive care. It redefines childcare as both a professional and societal priority, bridging the gap between technology and compassion.

In essence, NannyCare is more than a software system—it is a movement toward equity, dignity, and empowerment. By merging modern technology with empathy and education, it creates a future where mothers feel supported, nannies are respected as professionals, and children thrive in safe, caring, and stimulating environments. Through its vision and execution, NannyCare positions Ethiopia as a leader in digital solutions that uplift families, empower women, and nurture the next generation.

SheTrades

The She Trade's project is an innovative digital platform designed to empower women entrepreneurs by providing them with a space to showcase, market, and sell their products online. The goal of the platform is to bridge the digital divide and create economic opportunities for women, especially those in traditional and rural communities who may have limited access to digital markets. By connecting these women to a wider customer base through a simple and user-friendly platform, She Trade's seeks to promote financial independence, enhance women's participation in trade, and encourage local economic growth. Through this platform, women artisans, producers, and small business owners can display their goods, receive secure payments, and manage their orders seamlessly. The project envisions a future where women, regardless of their location or background, can compete fairly in the digital economy and benefit from the growing opportunities in e-commerce.

The idea for She Trade's stems from the recognition that many women possess exceptional entrepreneurial skills and produce high-quality local goods such as crafts, textiles, foods, and handmade products, but often lack the digital infrastructure to reach larger markets. Traditional barriers such as limited mobility, lack of digital literacy, and restricted access to banking services have historically hindered their participation in online trade. She Trade's aims to eliminate these barriers by offering a platform that is accessible, inclusive, and designed with simplicity in mind. It provides an intuitive interface where users can create profiles, list products with images and descriptions, track sales, and receive payments directly. By facilitating direct-to-consumer transactions, She Trade's helps reduce dependency on middlemen, ensuring that women earn fair value for their products.

The scope of She Trade's encompasses several key functionalities aimed at building a comprehensive and sustainable e-commerce ecosystem for women entrepreneurs. The platform will allow users to register as sellers, create and manage online storefronts, upload product details, and handle customer interactions. Buyers will be able to browse through various categories, add items to their carts, make secure payments, and track order delivery. To enhance trust and reliability, the system will incorporate features such as product reviews, seller ratings, and transaction histories. Moreover, She Trade's will integrate a secure payment gateway to support digital transactions, ensuring that sellers receive payments safely and promptly. The system will also include an administrative dashboard for monitoring activities, managing user accounts, and generating reports for performance evaluation. Future enhancements may include analytics tools to help women understand sales trends, training modules on digital marketing, and partnerships with logistics providers to streamline delivery processes.

From a technological standpoint, SheTrades is built using a robust and scalable stack that ensures both high performance and maintainability. The backend of the system is powered by Spring Boot, a Java-based framework that provides a strong foundation for building secure and reliable web applications. The frontend is developed using React, a powerful JavaScript library that enables the creation of dynamic, responsive, and user-friendly interfaces.

React allows for seamless interaction between users and the platform, ensuring a smooth browsing and shopping experience. For data management, the project utilizes PostgreSQL, a robust and open-source relational database system known for its scalability, reliability, and strong support for complex queries and transactions. This ensures that product listings, user information, and transaction data are securely stored and efficiently retrieved.

To enhance deployment and scalability, the application is containerized using Docker, which ensures consistency across different environments and simplifies deployment processes. Docker allows developers to package all necessary components into containers, reducing configuration issues and improving overall system performance.

In essence, SheTrades represents more than just a digital marketplace. It is a social and economic empowerment tool aimed at driving inclusiveness and gender equality in commerce. By leveraging modern technology to provide access, visibility, and opportunity, the project aspires to help women transition from local trading to global participation in the digital economy.